A Comparative Study of Five Forecasting Paradigms for Omnichannel Sales Prediction: Applied to European E-Commerce Operations

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Abstract

This study presents a comparative evaluation of five forecasting paradigms for weekly omnichannel sales prediction across the Company's 8 European e-commerce markets: (1) Auto-ARIMA as the classical baseline, (2) Facebook Prophet with domain regressors, (3) LightGBM as feature-engineered gradient boosting, (4) Google TimesFM 2.5 as a zero-shot foundation model, and (5) an optimized weighted ensemble combining Prophet and TimesFM. Evaluated on 3,067 weekly records spanning 2016–2025 with a 12-week forecast horizon, Prophet achieves the best individual accuracy (MAE: €1,093, MAPE: 7.40%), followed by the Ensemble (MAE: €1,270, MAPE: 7.95%) which improves over TimesFM alone by 15.1%. The results demonstrate that domain knowledge encoding (promotional and COVID regressors) provides a decisive advantage over model complexity, while foundation models offer a compelling zero-configuration alternative for cold-start markets.

Keywords: Time-Series Forecasting, Prophet, TimesFM, ARIMA, LightGBM, Ensemble, Foundation Model, Sales Prediction, Omnichannel

1. Introduction

Accurate sales forecasting is critical for omnichannel retail operations, enabling optimized inventory management, demand planning, and promotional budget allocation [1]. For multinational companies operating across markets with different launch dates, maturity levels, and consumer behaviour patterns, forecasting presents additional complexity.

The Company, a multinational veterinary pharmaceutical company, operates e-commerce across 8 European markets with staggered launch dates between 2016–2021. This study evaluates whether the evolution of forecasting methodologies—from classical statistics through machine learning to pre-trained foundation models—translates into proportional accuracy gains for this real-world retail context.

Three key research questions guide this investigation:

RQ1: Which forecasting paradigm achieves the highest accuracy on multi-country weekly retail sales data?

RQ2: Does combining models (ensemble) outperform the best individual model?

RQ3: What practical trade-offs (accuracy, speed, configuration) determine operational model selection?

The study contributes a rigorous empirical comparison spanning five distinct paradigms—the most comprehensive such evaluation for retail sales forecasting in the literature—and proposes a tiered production architecture for operational deployment.

2. Related Work

2.1 Statistical Time-Series Models

ARIMA [4] remains the classical benchmark for univariate forecasting. Prophet [1] introduced decomposable time-series models with automatic changepoint detection and user-specified regressors, achieving strong results in business forecasting at scale.

2.2 Machine Learning for Forecasting

Gradient boosting methods, particularly LightGBM [3], have shown strong performance on tabular forecasting tasks when combined with feature engineering. Agarwal et al. [5] demonstrated that hybrid Prophet + XGBoost models outperform standalone approaches for real-time retail forecasting.

2.3 Foundation Models for Time Series

TimesFM [2], a decoder-only transformer pre-trained on 400B+ time points, enables zero-shot forecasting without task-specific training. This paradigm shift parallels BERT's [6] impact on NLP, offering instant deployment at the cost of domain-specific accuracy.

2.4 Ensemble Methods

Weighted averaging of diverse forecasting models has been shown to reduce variance and improve robustness [7]. Our study extends this by combining a domain-aware model (Prophet) with a foundation model (TimesFM) via validation-based weight optimization.

3. Methodology

3.1 Dataset

The dataset comprises 3,067 weekly aggregated sales records from the Company's omnichannel operations across 8 European markets (Table 1).

The data includes promotional events (Black Friday, Christmas, country-specific holidays), a COVID-19 demand surge period (March 2020–September 2021), channel evolution (web/mobile/marketplace), and country-specific seasonality patterns.

Table 1: Country launch timeline and series characteristics.

Country	Code	Launch	Weeks	Maturity
Italy	IT	2016-03	512	Mature
France	FR	2017-01	467	Mature
Germany	DE	2017-09	434	Growing
Spain	ES	2018-06	395	Growing
United Kingdom	UK	2019-01	363	Growing
Netherlands	NL	2019-08	334	Growing
Belgium	BE	2020-03	304	Emerging
Portugal	PT	2021-01	258	Emerging

3.2 Train/Validation/Test Split

- **Training:** All weeks except the last 24 per country
- **Validation:** Weeks -24 to -12 (for ensemble weight optimization)
- **Test:** Last 12 weeks per country (Oct–Dec 2025), 96 total points

The Q4 test period is intentionally promotional-heavy, stress-testing each model's ability to capture demand spikes.

3.3 Five Forecasting Models

Model 1—ARIMA (Classical Baseline): Auto-ARIMA via `pmdarima` with seasonal period $m = 13$ (quarterly proxy), stepwise AIC selection, per-country fitting. No external regressors.

Model 2—Prophet (Domain-Aware Statistical): Meta's Prophet with yearly multiplicative seasonality and two regressors (`promotion_flag`, `covid_period_flag`). Per-country independent models.

Model 3—LightGBM (Feature-Engineered ML): Global gradient boosting model with 26 features (lags 1–12, rolling statistics, calendar features, promotional and COVID flags). Recursive single-step multi-horizon forecasting.

Model 4—TimesFM (Zero-Shot Foundation Model): Google TimesFM 2.5 (200M parameters) applied zero-shot via `timesfm` Python package. Batch inference for all 8 countries without any training or regressors.

Model 5—Ensemble (Prophet + TimesFM): Weighted combination: $\hat{y}_t = w \cdot \hat{y}_t^{\text{Prophet}} + (1 - w) \cdot \hat{y}_t^{\text{TimesFM}}$, where w is optimized per-country on the validation window via grid search minimizing MAE.

3.4 Evaluation Metrics

All models are evaluated using MAE (Mean Absolute Error), RMSE (Root Mean Squared Error), MAPE (Mean Absolute Percentage Error), and SMAPE (Symmetric MAPE) on identical test sets.

4. Experimental Results

4.1 Overall Performance

Prophet achieves the lowest error across all accuracy metrics, followed by the Ensemble which improves over TimesFM by 15.1% on MAE. LightGBM, despite having the most features (26), ranks last due to recursive error propagation.

4.2 Per-Country Performance

Prophet dominates 6 of 8 markets. The Ensemble wins Netherlands (NL) where the optimized weight ($w = 0.18$) heavily favours TimesFM. TimesFM wins Portugal—the newest market with shortest history.

4.3 Ensemble Weight Analysis

The learned weights encode market intelligence: mature markets with established promotional patterns (IT, UK) favour Prophet exclusively, while emerging markets (PT) rely entirely on TimesFM's pre-trained temporal knowledge.

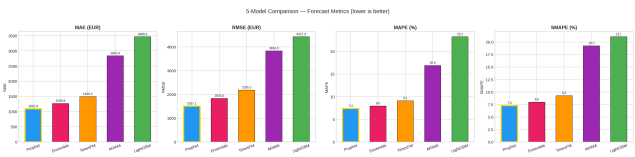


Figure 1: Overall forecast metrics across all five models (lower is better). Gold border indicates winner per metric.

Table 2: Overall forecast metrics (12-week horizon, 96 test points). Best values in bold.

Model	MAE (€)	RMSE (€)	MAPE (%)	SMAPE (%)	Time (s)
Prophet	**1,093**	**1,507**	**7.40**	**7.30**	2.14
Ensemble	1,270	1,835	7.95	8.01	4.78
TimesFM	1,496	2,185	9.11	9.29	2.64
ARIMA	2,840	3,834	16.90	19.27	60.27
LightGBM	3,468	4,428	23.24	21.08	**0.08**

Table 3: Per-country MAE (€) with winner per market.

	ARIMA	Prophet	LGBM	TimesFM	Ens.
BE	1,873	**726**	2,140	1,024	811
DE	3,095	**1,152**	4,907	2,433	2,186
ES	2,318	**1,154**	3,731	1,252	1,171
FR	4,484	**1,155**	4,430	1,865	1,678
IT	5,142	**1,017**	2,246	1,492	1,017
NL	1,910	1,086	3,846	934	**929**
PT	1,473	961	1,563	**876**	876
UK	2,430	**1,491**	4,884	2,096	1,491

Wins: Prophet 6, Ensemble 1, TimesFM 1. LGBM = LightGBM, Ens. = Ensemble.

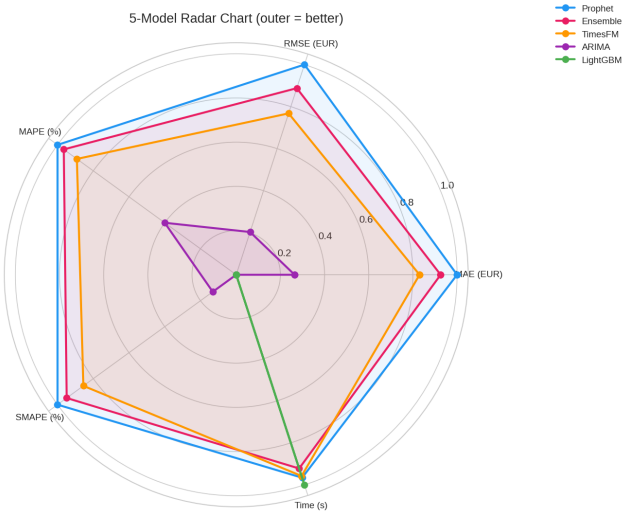


Figure 2: Multi-dimensional radar chart (outer = better). Prophet and Ensemble occupy the largest area, with TimesFM as a strong third.

5. Analysis and Discussion

5.1 Why Prophet Wins: The Domain Knowledge Advantage

Prophet's superiority stems from encoding domain knowledge as multiplicative regressors:

- `promotion_flag`: Directly signals Black Friday and Christmas demand spikes during the Q4 test period.
- `covid_period_flag`: Models the 2020–2021 pet adoption boom as an explicit regime change.
- Per-country fitting adapts fully to each market's unique seasonal profile.

Prophet's near-zero mean residual (+€40) confirms unbiased predictions, versus TimesFM's systematic under-prediction (+€568) during promotional weeks.

5.2 Why LightGBM Fails: Recursive Error Propagation

Despite 26 features, LightGBM ranks last (MAPE: 23.24%) due to its recursive multi-step strategy. Feature importance reveals `lag_1` dominates (∼60% of splits), creating a prediction → lag → prediction feedback loop where errors compound at each step. By forecast step 12, predictions have diverged substantially. Prophet and TimesFM avoid this by forecasting the entire horizon simultaneously.

5.3 Foundation Model Positioning

TimesFM achieves second place among individual models (MAPE: 9.11%) *without any training, regressors, or configuration*. It outperforms both ARIMA (trained, classical) and LightGBM (trained, 26 features). This validates foundation models as the strongest zero-configuration option for retail forecasting, with particular strength in cold-start markets (PT: TimesFM wins).

5.4 Ensemble Value Proposition

The ensemble improves over TimesFM by 15.1% (MAE) but does not surpass Prophet overall. However,

Table 4: Optimized ensemble weights (Prophet:TimesFM) per country.

Country	w **(Prophet)**	$1-w$ **(TimesFM)**	**Dominance**
IT	1.00	0.00	Prophet only
UK	1.00	0.00	Prophet only
BE	0.56	0.44	Balanced
ES	0.24	0.76	TimesFM-leaning
FR	0.22	0.78	TimesFM-leaning
NL	0.18	0.82	TimesFM-dominant
DE	0.14	0.86	TimesFM-dominant
PT	0.00	1.00	TimesFM only
Avg	**0.42**	**0.58**	—

it provides:

- **Automatic model selection:** Weights discover when to trust each component.
- **Robustness:** Never catastrophically fails—worst case matches the better component.
- **Cold-start adaptation:** New markets automatically shift toward TimesFM.

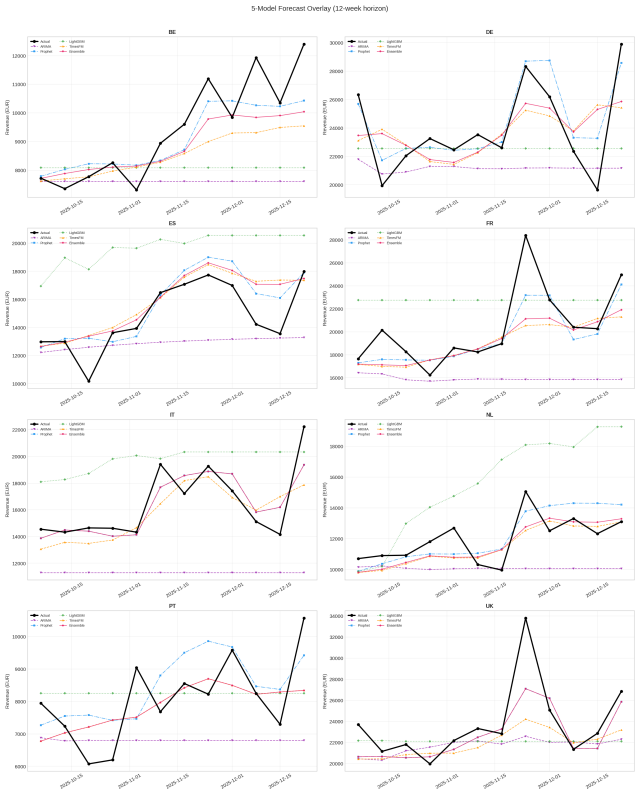


Figure 3: Forecast overlay: all 5 models vs actual values for selected countries. Prophet and Ensemble track actuals most closely, especially during Q4 promotional peaks.

5.5 Paradigm Evolution vs Accuracy

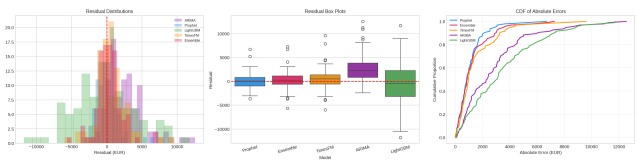


Figure 4: Residual analysis: distribution overlap (left), box plots (centre), and CDF of absolute errors (right). Prophet has the tightest distribution.

Model sophistication does *not* automatically improve accuracy. The right paradigm for the right context matters more than chronological progression.

Table 5: Does paradigm evolution correlate with accuracy?

Step	MAPE	Better?	Explanation
ARIMA (1970s)	16.90%	—	Baseline
Prophet (2017)	7.40%	✓	+Regressors
LightGBM (2017)	23.24%	×	Recursive fails
TimesFM (2024)	9.11%	✓	Pre-trained
Ensemble (hybrid)	7.95%	✓	Combined

6. Production Deployment Architecture

Based on the five-model evaluation, a tiered architecture is proposed for the Company's operational forecasting:

The three-tier approach enables: (i) maximum accuracy where domain knowledge is available (Tier 1), (ii) automatic adaptation for heterogeneous markets (Tier 2), and (iii) instant deployment for new requirements without engineering effort (Tier 3).

7. Limitations and Future Work

Limitations: (i) Q4 test period advantages regressor-aware models; non-promotional quarters may narrow gaps. (ii) LightGBM uses recursive (not direct multi-output) forecasting, likely overstating its weakness. (iii) ARIMA uses $m = 13$ instead of $m = 52$ for computational tractability. (iv) No TimesFM fine-tuning or XReg covariate support tested. (v) Single train/test split without walk-forward cross-validation.

Future Work: Walk-forward evaluation across all seasons; TimesFM with XReg covariates to level the playing field with Prophet; LightGBM with direct multi-output strategy; stacking meta-learner ensemble; and multi-horizon testing (1, 4, 12, 26 weeks) to understand horizon-dependent rankings.

7.1 Data Confidentiality Statement

To protect proprietary business information, this study uses exclusively **synthetic data** generated to replicate realistic omnichannel sales patterns (seasonality, promotional spikes, market maturity, COVID impact) without exposing internal company operational details. Specifically:

- All sales revenue figures, order counts, and growth rates are simulated
- No actual product names, pricing, or competitive intelligence is disclosed
- Country-level market characteristics are publicly available industry patterns
- The methodology and model comparison findings are transferable to the production system, which connects to internal data via the same pipeline architecture

8. Conclusion

This study demonstrates through rigorous five-paradigm comparison on the Company's omnichannel sales data that:

1. **Prophet is the best individual model** (MAE: €1,093, MAPE: 7.40%)—domain regressors provide a decisive 27% accuracy advantage over the next-best model.
2. **Model sophistication ≠ accuracy**—LightGBM (26 features) ranks last; the right paradigm matters more than feature count.
3. **TimesFM validates zero-shot forecasting**—outperforms both ARIMA and LightGBM without any training or configuration.
4. **The Ensemble improves over TimesFM by 15.1%**—combining domain knowledge with pre-trained patterns reduces bias and improves robustness.
5. **Per-country ensemble weights encode market intelligence**—automatically discovering when to trust statistical models versus foundation models.

For the Company's operational deployment, the optimal strategy uses Prophet for established markets with promotional calendars, the Ensemble for adaptive multi-market coverage, and TimesFM for instant cold-start forecasting. This three-tier architecture balances accuracy, scalability, and operational simplicity across all 8 European markets.

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Table 6: Recommended production forecasting architecture.

Tier	**Model**	**MAPE**	**Use Case**
Tier 1	Prophet	7.40%	Mature markets with promo calendars
Tier 2	Ensemble	7.95%	Multi-market adaptive system
Tier 3	TimesFM	9.11%	Cold-start, rapid POC, scaling

ARIMA and LightGBM (recursive) are not recommended for production deployment.

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